

Chapter 1

Ideals

1.1 Ideals

Definition 1. A subring A of a ring R is an (two-sided) *ideal* of R if for every $r \in R$ and every $a \in A$ then ra and ar are in A .

Example ?? is an ideal while example ?? is not an ideal.

Of course it will not be necessary to check the property “ ra and ar ” when the ring is commutative (we anyway in example ??). However keep in mind that the subring of a non-commutative ring may still be commutative –in which case the property must be checked.

Theorem 1. Ideal test. A nonempty subset I of a ring R is an ideal if and only if:

1. If $a, b \in I$ then $a - b \in I$.
2. If $r \in R$ and $a \in I$ then $ra \in I$ and $ar \in I$.

Proof. ($\Leftarrow$) Let I have properties 1 and 2. I is a ring by the subring test since property 2 is true for all $r \in I \subseteq R$, and property 1 is identical to the property 1 of the subring test. Property 2 also implies the absorbing property of ideals.

($\Rightarrow$) All ideals have property 1 since they are subrings, and property 2 by definition. $\square$

Let R be a commutative ring with unity¹, $c \in R$, and I the set of all multiples of c in R ($I = \{cr : r \in R\}$) Then I is an ideal.

Definition 2. The ideal in theorem 1 is called the *principal ideal generated by c* , denoted $\langle c \rangle$ ².

A *prime ideal* P of a commutative ring R is a proper ideal³ of R such that $a, b \in R$ and $ab \in P$ implies $a \in P$ or $b \in P$.

Definition 3. A *maximal ideal* M of a commutative ring R is a proper ideal such that whenever I is an ideal of R and $M \subseteq I \subseteq R$ then $I = M$ or $I = R$.

¹“with unity” is equivalent to “with identity”.

²Also often denoted (c)

³An ideal I in R is proper if $I \neq R$, i.e., I is a proper subset of R .

1.2 Congruence and ideals

We are familiar with the idea of congruence. Ideals can be congruent as well.

Definition 4. Let I be an ideal in a ring R and let $a, b \in R$. Then $a \equiv b \pmod{I}$ if $a - b \in I$. We say “ a is congruent to b modulo I ”.

Theorem 2. Let I be an ideal in the ring R . Then congruence modulo I in R is an equivalence relation.

Let I be an ideal in the ring R . If $a \equiv b \pmod{I}$ and $c \equiv d \pmod{I}$, then

1. $a + c \equiv b + d \pmod{I}$
2. $ac \equiv bd \pmod{I}$

1.3 Cosets

Recall that the elements in $\mathbb{Z}$ each fell into n classes of $\mathbb{Z}_n$ denoted $[a]$ for an $a \in \mathbb{Z}_6$. For $\mathbb{Z}_6$, $2 \equiv 8 \pmod{6}$ so then $8 \in [2]$. We can easily find all the elements in $[2]$ by repeatedly adding 6 (since we are working with modulo 6). Similarly we could find all the elements of each class.

$$\begin{aligned}
 [2] &= \{\dots, -2, 0, 2, 8, 14, \dots\} \\
 [3] &= \{\dots, -3, 0, 3, 6, 15, \dots\} \\
 &\vdots \\
 [a] &= \{\dots, a - n, a, a + n, a + 2n, \dots\} \\
 &\vdots \\
 [6] &= \{\dots, -6, 0, 6, 12, \dots\}
 \end{aligned}$$

The elements of each class of $[a]$ can each be expressed as $a + kn$ for some $k \in \mathbb{Z}$. It thus would make sense to write $a + 6\mathbb{Z}$ instead of $[a]$. This is in fact what we will do when expressing the congruence class of elements in a ring modulo an ideal.

Definition 5. Let I be an ideal in R and $a \in R$, then the *congruence class of a modulo I* is the set of all elements of R that are congruent to a modulo I , i.e.,

$$\begin{aligned}
 \{b \in R : b \equiv a \pmod{I}\} &= \{b \in R : b - a \in I\} \\
 &= \{b \in R : b = a + i \text{ for } i \in I\} \\
 &= \{a + i : i \in I\}
 \end{aligned}$$

The congruence class of a modulo I will be denoted $a + I$ (instead of $[a]$). The congruence class $a + I$ is also called the (left) *coset of I in R* .

Definition 6. If I is an ideal in the ring R , then the set of all cosets of I is called the *factor ring*⁴ R modulo I and is denoted R/I .

Theorem 3. Let I be an ideal in the ring R and let $a, c \in R$. Then $a \equiv c \pmod{I}$ if and only if $a + I = c + I$.

Proof. ($\Rightarrow$) Suppose that $a \equiv c \pmod{I}$. First we will show that $a + I \subseteq c + I$. Let $b \in a + I$. By definition, $b \equiv a \pmod{I}$. Since $a \equiv c \pmod{I}$ we know that $b \equiv c \pmod{I}$ by transitivity

⁴Also called the *quotient ring*.

(theorem 2). Thus $a + I \subseteq c + I$. Similarly we can show that $c + I \subseteq a + I$ since $a \equiv c \pmod I \Rightarrow c \equiv a \pmod I$.

($\Leftarrow$) Suppose that $a + I = c + I$. Since $a \equiv a \pmod I$ by reflexivity (theorem 2), we have that $a \in a + I$ and thus $a \in c + I$. By definition we then have that $a \equiv c \pmod I$. $\square$

Corollary 1. *Let I be an ideal of the ring R . Any two cosets of I are either disjoint or identical.*

Proof. Suppose that $a + I$ and $b + I$ are not disjoint. Then there is an element $r \in R$ such that $r \in a + I$ and $r \in b + I$. So then $r \equiv a \pmod I$ and $r \equiv b \pmod I$. By symmetry $r \equiv b \pmod I \Rightarrow b \equiv r \pmod I$. Then by transitivity $a \equiv b \pmod I$. By the theorem above, we have that $a + I = b + I$. $\square$